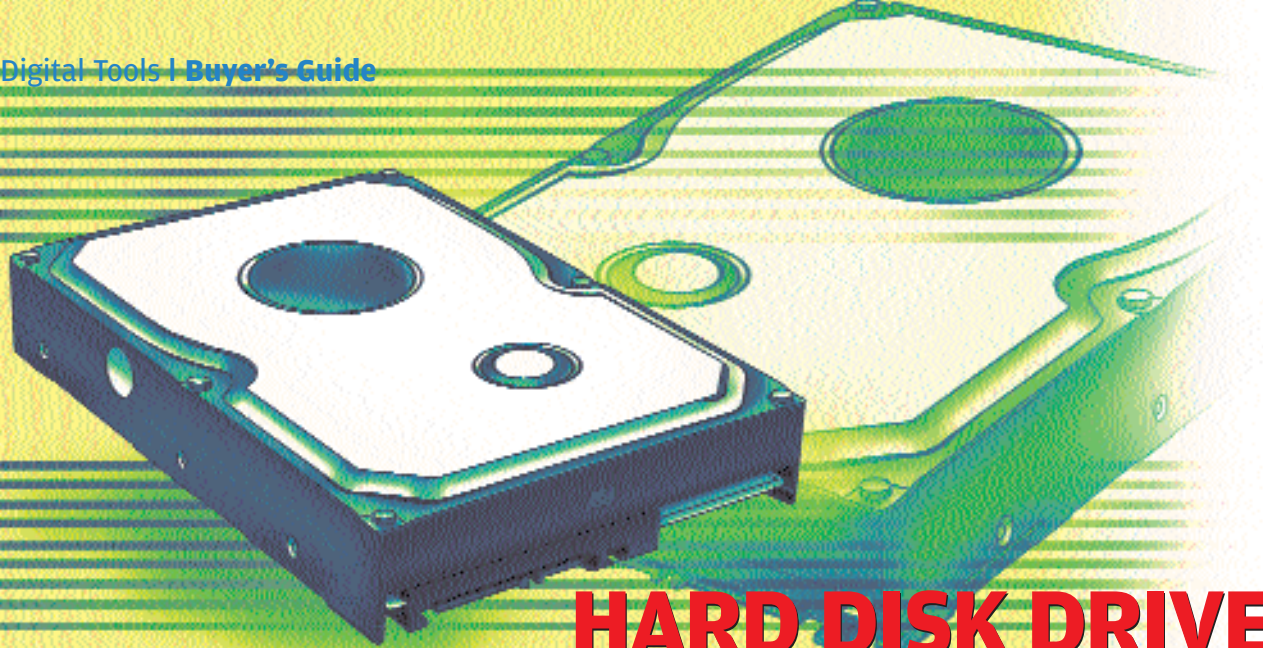




HARD DISK DRIVES

Those who have been associated with computers for the past decade would recollect the 32 GB limit—an indication of how early motherboard manufacturers underestimated the development of storage media. Since then, storage capacities seem to have skyrocketed; even the speed of data transfer has increased steadily. The interfacing technology between the motherboard and the disk drive has seen a major improvement as well.

Questions to ask the dealer

What models do you have?

Although you have a model in mind, you should ask the dealer for any new models that have arrived, and compare the features with the one you've decided on.

Is the drive brand new?

Don't let the dealer con you with a used product. Many users are not aware of bad sectors and they use the drive with the defect—some dealers may try their luck. So check the disk at the dealer's outlet by attaching it as a slave to their test machine and see that the drive is unpar-

tioned. If possible, perform a disk check using a drive-check utility.

What is the warranty type?

Prefer drives that come with a replacement warranty.

Interesting Facts

IBM's 'Drum' epitomises the amount of research that has gone into developing what is probably the most important component of a PC—the hard disk. You can replace your processor; all it takes is money. But if your hard disk were to crash, what happens to your data?

IBM's Drum was a rotating cylindrical device on which data was stored as magnetic patterns. It was replaced with hard drives that resemble the ones we use today, except that they had much lower capacities and slower speeds.

Usage Tips

Internal hard drives are fragile and are not designed to be repeatedly attached and detached from a system. It is advisable that you be extremely careful when transporting your hard drive.

Do not keep the hard drive loose or vertical within your cabinet. Tightly screw in the drive to the cabinet, and do not cover it with any insulating material such as plastic. A hard drive, like most other components in a PC, needs to dissipate heat and needs ventilation.



Only use cables that are recommended by the manufacturer to interface the hard drive with the motherboard. For instance, if an ATA133 hard drive is interfaced using an ATA66 IDE cable, data transfer speeds are affected.

If you are using Windows, an occasional disk defragmentation would not only clear cluttered data, but will also improve the performance of the operating system.

The Future

The growing need for space and speed, especially at server levels, has brought researchers back to the drawing board. The need of the moment is, quite simply, higher data densities.

The single most exciting development we see happening is Perpendicular Recording (PR). PR offers much higher storage capacities. In PR, data bits will be stored "one above the other" as a stack of bits, rather than "side-by-side" as they are today. We're keen to get our hands on a drive that uses this technology—and we're sure you are, too! ■



Points To Ponder

How large a hard drive do I need?

A good way to decide this is to see the amount of times you have burnt CDs to shift data off your drive. If you do not have a CD-Writer, try and remember how many times you wished you had one, and buy a hard drive accordingly. This question, though, is often decided more by the budget than by space requirements. Still, if you can afford it, buy the largest hard drive you can afford—you won't regret it!

What do I look for in a hard drive?

Currently, hard drives are available in two different interfaces—Parallel ATA (PATA) and Serial ATA (SATA). If possible, buy the newer SATA drives, provided your motherboard supports these, of course. SATA drives are faster than PATA, and are generally available

in much larger capacities—as much as 500 GB. The rpm factor is very important as well, as is the buffer/cache size. Look for a hard disk with a buffer/cache size of 8 MB or more, and an rpm speed of at least 7200.

What type of hard disk will my system support?

Understanding your system is the most important requirement, because if you buy a SATA hard drive and your system doesn't support it, you will be forced to upgrade your motherboard or exchange your drive for an older PATA one.

Read through your motherboard manual for SATA support, or open up the cabinet and look for the model number and then search the Internet for the model specifications.